Exploring Digital Rights Management in Music Streaming Platforms:
The Case of Spotify

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November 2025

Abstract

This literature review critically investigates Digital Rights Management (DRM) within music streaming platforms, focusing on Spotify as a primary case. It includes a systematic synthesis of scholarly research examining Spotify's DRM systems, business models, licensing frameworks, artist compensation mechanisms, user rights, and the accompanying ethical and legal challenges. The review explores how DRM shapes music ownership, controls user access, and impacts artist autonomy within the digital ecosystem. Core findings reveal that Spotify's DRM curbs piracy and automates royalty payments globally but also concentrates platform power, limits creative control, and raises ethical issues concerning fairness, transparency, privacy, and inclusivity. The review identifies significant regulatory and audit gaps, particularly in royalty transparency and data privacy protections. Drawing on the ACS Code of Ethics and information security principles, the analysis calls for reforms prioritizing ethical governance, enhanced fairness, and broader public interest. These changes are essential for realigning digital music distribution toward greater transparency, equity, and accessible participation for all stakeholders, from artists to end users.Srivastava, 2022Zhang and Lou, 2011

Keywords: Digital Rights Management, Spotify, Copyright, Artist Rights, User Access, Information Security, ACS Ethics

1 Introduction

1.1 Background and Context

Contemporary music distribution has shifted dramatically from physical media (such as CDs and tapes) to a widespread streaming model driven by advanced Digital Rights Management (DRM) systems. These systems enable platforms like Spotify the world's most influential music streaming service—to secure digital content, automate royalty calculations, and prevent unauthorized sharing through robust licensing, encryption, and monitoring mechanisms. Because DRM sits at the intersection of information security, digital economics, copyright law, and ethical governance, studying its implementation within Spotify provides a meaningful lens for understanding how digital platforms reshape relationships among artists, users, and rights holders.

I chose this topic because DRM is now central to global music distribution and raises important questions about privacy, fairness, transparency, and platform power—issues that are highly relevant to contemporary information security practice and policy. Spotify's dominant role in the industry makes it an especially valuable case study for analysing how technical controls influence economic outcomes, user rights, and creator autonomy.

To build the foundation for this review, I conducted a broad search on Google Scholar and initially identified 48 relevant articles. Of these, 17 were shortlisted based on their non-technical style and their direct engagement with business, legal, ethical, or structural issues in DRM systems.As final selection, eight key articles were selected for in-depth analysis because they provided focused insights on Spotify or examined DRM practices closely aligned with its operational model. This systematic screening process allowed the review to draw on literature that reflects both scholarly depth and practical relevance to Spotify's DRM ecosystem.

1.2 Relevance to Information Security

DRM organizes permission-based access, a key information security challenge, and protects digital assets by combining cryptography, licensing frameworks, and contractual governance. In globally interconnected platforms, it influences digital trust, data privacy, surveillance, and compliance. Rafi et al., 2023Beekhuyzen, 2006

1.3 Research Aim and Questions

The aim of this literature review is to analyse the effect of DRM on the relationships and power dynamic between artists, users , copyright holders and platform in the Spotify ecosystem. Research Questions include:

1. How does Spotify's use of DRM redefine ownership, licensing, and access in music distribution?
2. What are the equity, transparency, and privacy impacts for users and artists?
3. Which legal and ethical challenges arise, and what reforms are emerging?

1.4 Article Selection Rationale

The topic was chosen for its intersection of copyright, technology, ethics, and market practice in digital music. Of the 48 relevant articles identified, 17 were selected for deeper screening due to their non-technical style and direct engagement with legal, economic, or ethical facets. The final 8 include:

1. **Nick Scharf, "The evolution and consequences of digital rights management in relation to online music streaming" :**This article is chosen because it provides a comprehensive, critical analysis of how DRM has evolved in online music streaming and explicitly examines Spotify's role as a leading example. It links legal and policy changes with the practical impacts of DRM on users and artists, focusing on end-user license agreements, ownership, and market power. Its clear focus on Spotify's business model and its implications for copyright, licensing, and user rights makes it highly relevant for DRM and streaming researchScharf, 2022
2. **Ali Matin, "Digital Rights Management in Online Music Stores" (Article2):** This article was selected because it provides an in-depth legal and practical analysis of how DRM and the Digital Millennium Copyright Act (DMCA) have shaped—and often negatively impacted—online music distribution. It offers clear insights into the unintended consequences of heavy-handed copyright law, the balance between copyright protection and consumer rights, and backs its arguments with real-world industry examples. The article's focus on both legal and business implications makes it highly valuable for examining the intersections of information security, law, and digital media.Matin, 2007
3. **Jenine Beekhuyzen,"Digital Rights Management and the Online Music Experience:"**This article was selected because it provides rare, in-depth empirical research on how users and musicians actually experience and interact with online music systems, using interviews, observations, and focus groups to analyze both authorized and unauthorized access methods. Its value lies in exploring real online music communities, digital rights management (DRM) restrictions, and usability issues—directly relating to the overall online music experience and highlighting how

DRM technology (including that of platforms like iTunes and, by extension, systems like Spotify) shapes user opportunities and challenges. The insights are highly relevant for understanding social, technical, and ethical implications of DRM in modern online music environments, and the analysis helps inform better system design and policy in a context increasingly dominated by services like Spotify. Beekhuyzen, 2006

4. **Amir Rafi, Carlton Shepherd, and Konstantinos Markantonakis ,"A First Look at Digital Rights Management Systems for Secure Mobile Content Delivery":** This article was selected because it offers the first comprehensive security evaluation of leading mobile DRM systems—Google Widevine, Apple FairPlay, and Microsoft PlayReady. It systematically compares their technical designs, strengths, and vulnerabilities, specifically highlighting how these DRM technologies are deployed by streaming platforms (including Spotify). The paper is highly relevant due to its focus on the security of mobile content delivery, a crucial topic for online media services, and provides actionable insights into both existing weaknesses and potential future improvements for securing digital content in a mobile, cloud-based landscape.Rafi et al., 2023

5. **S.R. Subramanya and Byung K. Yi,"Digital rights management"** This article was selected because it clearly explains the architecture, policies, and core mechanisms underpinning digital rights management (DRM) systems from both a technical and broader digital content ecosystem perspective. Its value lies in offering a detailed yet accessible overview of DRM's major components, functionalities, and challenges—including those relating to secure content delivery, user rights, privacy, and system interoperability. This foundational review makes it especially useful for understanding the basic principles and big-picture context of DRM, which supports further research or case study analysis in areas like online music and video streaming.Subramanya and Yi, 2006

6. **Prashansa Srivastava,"Digital Rights Management: The New Copyrights":** This dissertation was selected because it offers a comprehensive legal and technological analysis of digital rights management (DRM) with a particular focus on global frameworks, legislative developments, and real-world legal cases. The work thoroughly explores DRM's historical evolution, privacy concerns, copyright enforcement challenges, and jurisdictional issues—essential for understanding both national and international perspectives. While it does not mention Spotify specifically, the broad coverage of online copyright, fair use in the digital domain, collective management, and digital content regulation is directly relevant to platforms like Spotify, which rely on DRM for streaming media protection. This makes the dis-

sertation especially useful for foundational research on DRM laws and the modern digital media landscape.Srivastava, 2022

7. **Jizhong Zhao, Yong Qi, and Zhaofeng Ma,"Secure Multimedia Streaming with Trusted Digital Rights Management"** This article was selected because it presents an innovative approach to multimedia DRM by proposing a partial encryption method to protect streaming and downloadable digital content in real time. It offers a well-structured technical solution that balances security, usability, and efficiency, addressing major industry needs for copyright protection and rights management. While this article discusses the general principles and architecture of DRM systems (such as those adopted in commercial music and video streaming), it does not specifically reference Spotify by name. However, the methods analyzed are directly relevant for large-scale streaming platforms, including Spotify, that require efficient, scalable, and secure DRM for their content delivery networks.Zhao et al., 2005

8. **Jin Zhang and Wei Lou,"The Digital Rights Management Game in Peer-to-Peer Streaming Systems":** This article was selected because it rigorously models digital rights management (DRM) within peer-to-peer streaming systems using game theory, presenting new strategies to balance content provider and user interests, address misbehaviors (like freeriding and jailbreaking), and optimize DRM deployment. The work is significant in the context of real-time streaming economics, incentives, and security. While Spotify is not explicitly discussed, the peer-to-peer and streaming content dynamics highlighted here are relevant to understanding scalable DRM solutions for modern platforms like Spotify, which require robust content protection yet must balance network efficiency and user experience in large-scale streaming environments.Zhang and Lou, 2011

1.5 Structure of Review

The review proceeds from the business and legal foundations of DRM—including platform ownership, licensing, and copyright frameworks—then examines Spotify's technical design and DRM workflow, detailing encryption and access controls, contract management, and piracy prevention measures. This is followed by an in-depth analysis of artist and user rights, exploring the effects of DRM on compensation, autonomy, auditing, ownership, and fair use. Ethical and governance considerations are then addressed, including transparency, inclusion, equity, accountability, and the ethical application of standards like the ACS Code. The review concludes with recommendations for building more transparent, accessible, and equitable DRM designs for music streaming platforms like Spotify

2 Business Models and Company Ownership of DRM

2.1 Centralisation and Enforcement of Access

Music streaming platforms such as Spotify have transformed music distribution by moving from physical and digital ownership models to cloud-based access, which is fundamentally supported by digital rights management (DRM) systems. DRM allows platforms to offer "permission-based" access, which shifts the product offered from a privately owned copy to a bundled series of access and usage licences. DRM effectively restricts what users can do with music such as reselling, copying, sharing, or transferring tracks illegally and this sometimes creates tension with longstanding copyright principles concerning ownership and the exhaustion of rights. Nevertheless, the dominance of streaming services has contributed to a decline in piracy, as users are increasingly channelled toward legally protected alternatives supported by robust DRM ecosystems. Scharf, 2022 Matin, 2007
Even though record labels and in some cases, artists retain copyright, platforms like Spotify further centralise business control by operating and owning the DRM infrastructure. Spotify's servers encrypt all tracks and authorise playback only through tightly controlled mechanisms. This form of technical possession means that users never receive the original audio files; instead, all access is granted through DRM-managed licences. This model gives Spotify decisive power over where, when, and how streaming occurs by enforcing territorial, temporal, and device-based restrictions defined in licensing agreements with record labels. When a user's subscription ends, they can no longer access the content, and even previously downloaded songs become inaccessible. The licences granted to users are non-exclusive, non-transferable, and revocable, ensuring Spotify maintains enduring control over its music catalogue. Scharf, 2022 Matin, 2007

2.1.1 Synthesis

The literature broadly agrees that DRM shifts music consumption from ownership toward platform-controlled access, though scholars differ on whether this shift primarily safeguards content or strengthens corporate control. A key gap identified across studies concerns how this erosion of ownership affects user autonomy and long-term cultural access. For Spotify, this dynamic is especially significant because it enables strong user lock-in and reinforces its dominant position within global music distribution ecosystems.

2.2 Licensing and Rights Possession

Record labels such as Universal, Sony, Warner, and other rights holders maintain copyright over their music content. However, Spotify obtains licences to distribute this content under contractual terms that dictate how the music can be played, on which devices, and in which regions it can be accessed. Users can only interact with DRM-protected versions of the content and never receive the original audio files because Spotify's DRM servers encrypt and store all tracks securely. This structure places the entire DRM infrastructure—and the controls governing encryption, playback, and distribution under Spotify's proprietary management, granting the platform significant gatekeeping power over user access. Rafi et al., 2023

2.2.1 Synthesis

Across the literature, researchers consistently recognise that artists and rights holders retain copyright, while platforms control access through DRM. However, debate persists regarding whether this arrangement supports creators or reinforces platform-level dominance. A recurring gap involves how independent artists experience these licensing structures compared with major labels. This tension between legal ownership and practical control highlights ongoing concerns about how DRM governs access and participation within Spotify's distribution model.

2.3 Licensing and Royalty Frameworks

Music streaming platforms like Spotify employ licensing agreements with major record labels, freelance artists, and copyright holders, using Digital Rights Management (DRM) to prevent unauthorized access and monitor playback restrictions. Typically, these contracts are performance-based and specify royalty payments determined by subscription tier, region, and the number of user streams.

Scharf, 2022Matin, 2007 DRM is the main pivot of this ecosystem, and it makes sure that every playback is monitored and logged, verifying when and where content is accessed and guaranteeing compliance with contractual terms. This arrangement supports the confidence of rights holders and platforms that usage is accurately measured and compensated, because this system enables precise and automated royalty payments and supports regulatory compliance.

Scharf, 2022Matin, 2007 The DRM framework also supports the enforcement of limited fair use, occasionally allowing certain user freedoms within tightly controlled commercial terms. However, the main aim still remains exclusivity, as DRM restricts music from being played, copied, or moved outside the Spotify platform. These restrictions prevent the formation of independent secondary markets or illegal music streaming platforms and maintain tight control over digital music distribution. As a result, this system supports market expansion and has contributed significantly to the reduction of piracy, with users being converted to legitimate, easily accessible alternatives.Srivastava, 2022

2.3.1 Synthesis

Studies generally indicate that DRM improves the accuracy of royalty tracking, yet issues related to opaque payment models and limited algorithmic transparency remain unresolved. Scholars continue to debate whether DRM promotes equitable compensation or amplifies existing inequalities among artists. The absence of independent audits within DRM-based royalty systems represents a significant weakness in current research. For Spotify, transparent and accurate royalty reporting remains critical to sustaining artist trust and maintaining platform credibility.

2.4 Spotify's Business Strategy with DRM

Spotify leverages DRM as a core business strategy to differentiate its services and generate profit by monetising its offerings through a tiered subscription system. Free users have limited access, cannot use all features, and are unable to download music for offline use, whereas premium subscribers enjoy unrestricted, high-quality playback and the ability to download tracks within the DRM-protected ecosystem. However, all consumption whether streamed online or played offline is confined to authorised Spotify applications or devices, ensuring that songs and playlists cannot be shared or transferred freely beyond the platform's control. By applying these restrictions, Spotify not only protects against unauthorized duplication but also strengthens platform exclusivity and reinforces user lock-in to its services.Scharf, 2022 Rafi et al., 2023

The most important aspect is that proprietary control of its DRM infrastructure gives Spotify considerable bargaining power with rights holders and advertisers. Being in charge, all content delivery through technical and contractual controls allows Spotify to present enhanced value to advertisers and copyright owners, with all user access being logged and monitored. This system supports Spotify's capacity for rapid innovation in pricing and service models while aligning commercial obligations and maintaining competitive market advantages.Matin, 2007

2.4.1 Synthesis

There is broad scholarly agreement that DRM-based restrictions reinforce platform lock-in by limiting file transfers and authorised playback. However, opinions diverge on whether these restrictions represent excessive constraints for consumers or reasonable commercial practices. A persistent gap concerns the long-term impact of subscription-based DRM models on user freedom and market competition. Within Spotify's context, DRM functions not only as a protective mechanism but also as a central business strategy underpinning revenue, pricing tiers, and competitive advantage.

2.5 Artist Protection, Revenue Integrity and Ethical Tensions

DRM-based analytics on Spotify offer robust protection against unauthorized duplication, piracy, and manipulation of royalty-related metadata for artists and copyright holders. Royalty frameworks are tied to DRM-generated data and help rights holders monitor whether their work has reached global audiences through platform dashboards and reports. Nevertheless, clarity concerns persist, as many contractual terms remain uncertain, and the power over reporting and verification has largely remained with Spotify and major music labels, leading to ongoing debates about equity and fairness for artists.Rafi et al., 2023 Scharf, 2022

Tensions related to ethics frequently emerge from this business model, as DRM centralises significant cultural and economic authority in the hands of a few major platforms and rights holders. This raises challenges concerning long-standing ideas of ownership, user rights, and creative freedom. While DRM can enforce certain fair-use provisions, its main function remains exclusion, as music can only be played under tightly defined authorisation. This can compromise diversity, access, and the evolution of independent or decentralised secondary markets. As a result, the licensing environment becomes increasingly integrated, where user rights are replaced by a permission-based model, and market competition becomes constrained as platforms consolidate their authority.Srivastava, 2022 Scharf, 2022Matin, 2007

This dynamic also raises important questions about public interest. While DRM systems protect the economic interests of music creators and distributors, they must be continuously examined to ensure that protective measures do not unnecessarily restrict cultural participation or undermine access to cultural works, especially for marginalised communities or small independent artists. In response, international courts and regulators have begun to recognise these tensions and are exploring new frameworks and exceptions such as adjustments to fair use and fair dealing to achieve a better balance between protection, innovation, and public access in the digital era.Srivastava, 2022

2.5.1 Synthesis

The literature acknowledges that DRM protects artists through precise metadata tracking and anti-piracy features. Nevertheless, numerous studies highlight ethical concerns such as opaque algorithmic processes, restricted auditability, and the concentration of decision-making power within major platforms. A notable research gap involves understanding how these systems affect smaller or independent artists who may not benefit equally from DRM-enabled structures. These issues are particularly relevant to Spotify, where perceptions of fairness and transparency influence artist satisfaction and long-term platform relationships.

3 DRM and Copyright Challenges in the Spotify Ecosystem

3.1 Eorison of User Rights and Market Ownership

The use of digital rights management (DRM) in Spotify represents an essential shift in the way music is distributed, marking a transformation from ownership-based models such as CDs, tapes, and digital downloads to a permission-based license ecosystem. It is not like traditional copyright models grounded in ownership and the idea of exhaustion, which permit secondary markets and resale. Instead, music streaming through Spotify provides only limited, revocable licenses to the user, tightly regulating the terms of access. Although copyright is important, the foundation for contracts and DRM enforcement applies only to Spotify, while the music copyright is still owned by labels and artists. This means that users are granted only non-transferable licenses rather than actual ownership of digital copies.Scharf, 2022Matin, 2007

The complexity and lack of clarity in end-user license agreements intensify this trend, creating a challenging relationship between the industry, stakeholders, and users. Typically, licenses and agreements are non-negotiable, limiting the ability to transfer ownership as well as restricting secondary markets and user flexibility. Both US and EU jurisdictions define this shift as emblematic of broader legal and technological transformations, where restrictive DRM and standard licensing replace direct music ownership and threaten competitive diversity.Scharf, 2022

3.2 High-Profile Copyright Disputes: The Taylor Swift catalogue Removal

The most popular case that happened in Spotify is the removal of Taylor Swift's music catalogue in 2014, which became one of the most prominent DRM and copyright debates. This unexpected move, brought about by Swift and her management, raised concerns that Spotify's licensing and royalty framework immersed in DRM infrastructure threatened both fair compensation and creative agency for artists. The withdrawal of Swift's catalogue was a protest against streaming's perceived devaluation of music and highlighted the limitations artists face in controlling their work once licensed to major platforms.

So this case highlights the tension created by DRM, where DRM platforms can track, restrict, and revoke access at any time. Although it can serve the interests of copyright holders, it can also work to the disadvantage of artists as well as end users. Artists who rely on platform-based royalties lack the leverage to negotiate more favourable terms or to ensure their content's continued availability on platforms, emphasizing how DRM business models amplify the market power of platforms and major labels at the expense of creators. Scharf, 2022

3.2.1 Synthesis

Researchers generally agree that DRM transforms music from an ownership model to a service-based access model, yet they differ on whether this transformation benefits users or limits market diversity. Persistent conflicts remain between DRM practices and traditional copyright principles, with little research on broader long-term legal implications. High-profile cases, such as the removal of Taylor Swift's catalogue, emphasise ongoing tensions related to artist control. For Spotify, these challenges reinforce the need to balance operational efficiency with policies that support artists and users more equitably.
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4 Security and Incidents and Cyberattacks related to DRM/Spotify

4.1 DRM System Vulnerabilities and Threats

The legal protection received by DRM technologies has often hindered independent academic and security research, but it occasionally masks vulnerabilities in practice. In recent years, numerous threats to DRM systems have been identified; for example, proprietary DRM protocols such as Google Widevine, which is widely used by multiple major streaming platforms including Spotify and video streaming services such as Netflix, Disney+ , HBO Max and Amazon video, have been shown to carry vulnerabilities like cache exposure, poor policy management, and failures in key management. So these flaws can be weaponized by both end users and malicious actors to strip DRM protections or exploit user data, raising the stakes for both copyright protection and user privacy.Rafi et al., 2023

For cyberattacks in DRM systems, it has been pointed out that DRM cannot resist attacks such as man-in-the-middle exploits, side-channel attacks, and server-based breaches. While public breaches on Spotify are uncommon due to legal restrictions on reverse engineering, researchers continue to identify recurring flaws in other cloud-based DRM architectures, so the broader digital environment becomes clearly vulnerable.Rafi et al., 2023

4.2 Songwriters' Royalty Disputes: Spotify vs Kobalt

Debates over royalties are native to streaming platforms, and Kobalt, as a high-profile publishing entity, has also been involved. Royalty tracking is tightly consolidated through DRM, as each play is logged, tagged, and accounted for via algorithms and contractual infrastructures. In application, however, conflicts in data tracking, contractual ambiguities, and delays in royalty payments have led to widespread tension between Spotify, music publishers, and songwriters. These issues boost artists' dependence on platform-controlled systems for validation in economic and market participation.Scharf, 2022

4.3 Conclusion for DRM and Copyright Challenges in the Spotify Ecosystem

Problems of DRM and copyright in the Spotify ecosystem reflect a broader realignment of legal, technical, and market standards, with platforms using DRM not only to combat piracy but also to cement their central role as cultural gatekeepers, which sometimes causes detriment to artists, consumers, and market competition. Well-known incidents such as the Taylor Swift catalogue removal are symbolic of the complex, often adversarial interaction between creators and streaming platforms, emphasizing the urgent need for transparent, balanced frameworks that sufficiently address the interests of all stakeholders in the digital music industry.

4.4 Synthesis

Although legal constraints often limit the depth of security research, scholars recognise that DRM systems contain technical vulnerabilities. Existing studies predominantly focus on video-streaming DRM such as Widevine, resulting in a lack of research specifically examining risks in music-streaming DRM architectures. This gap limits understanding of how cyberattacks may affect the integrity and reliability of Spotify's platform. Addressing these issues is essential for ensuring secure content protection, safeguarding user data, and maintaining compliance with contractual obligations.

5 Artist Rights and Access Control

5.1 Artist Access to DRM data

Spotify's DRM-based analytics infrastructure transforms how artists interact with listener data. The dashboard of Spotify for Artists presents artists and rights holders with access to granular metrics derived from tracking technologies included in DRM systems, such as counting the number of plays, the region of listeners, platform usage timings, and more. DRM guarantees that this data is linked to accurate metadata for each track, adding clear ownership attribution and reducing misidentification, which was known as an issue that plagued music reporting in the pre-streaming era. Furthermore, by revealing which territories and demographics engage most heavily with their work, artists can target marketing spend and plan licensing strategies with unmatched precision. These analytics offer empirical proof of market strength, regional popularity, and long-term trends, which further supports direct negotiations or licensing discussions. Beekhuyzen, 2006 Srivastava, 2022Scharf, 2022

5.2 Benefits for Musicians

1. **Reliable Revenue Tracking** – DRM integrates with international payment infrastructures, applies contractual splits in real time, and permits the automated logging of all streams. This makes it easier for even independent musicians to access worldwide royalties by minimizing the expensive delays and mistakes associated with manual reporting.Beekhuyzen, 2006 Subramanya and Yi, 2006

2. **Piracy Prevention** – Streaming DRM prevents unauthorized duplication by encrypting files and permitting playback strictly through authenticated clients such as the Spotify application, legal streaming platforms, and controlled access whenever contract terms or payments fail. By doing this, it helps protect artists from having their music leaked before release or before commercial distribution, allowing artists to maintain control over their careers.Matin, 2007Zhang and Lou, 2011Zhao et al., 2005

3. **Brand Control and Global Reach** – DRM-secured distribution ensures that artists' catalogues cannot be repackaged, altered, or redistributed without permission, preserving both the artistic intent and the commercial image of the artist worldwide. It also supports simultaneous global release routines, levelling the playing field for audiences everywhere.Subramanya and Yi, 2006Rafi et al., 2023

5.3 Challenging and Ethical Concerns

1. **Royalty Transparency Issues** – Although DRM gathers comprehensive performance data, artists rarely receive independent audits or breakdowns of deductions, splits, or algorithmic adjustments, and the algorithms that convert streams into payments are still opaque. This lack of openness has resulted in ongoing conflicts and even well-publicized catalogue withdrawals by well-known performers. Srivastava, 2022Matin, 2007

2. **Power Asymmetry and Platform Dependence** – Artists or labels must accept standardized contract agreements, but these contracts on Spotify are non-negotiable, leaving new and independent acts at a disadvantage compared to large rights holders. Increased reliance on platforms and the potential for creative output to be influenced by platform algorithms are two consequences of concentrating

technical control in the hands of a small number of platforms.Subramanya and Yi, 2006

3. **Fairness and Equity (ACS Principle)** – According to the ACS Code and numerous researchers, DRM demands genuine fairness, not just reliable tracking, but also open algorithmic practices and the option to contest or audit platform results. By design or inertia, the current system frequently violates these values, which could discourage innovation and erode confidence. Srivastava, 2022Rafi et al., 2023Beekhuyzen, 2006

5.4 Synthesis

Researchers widely agree that DRM-based analytics provide artists with valuable insights into audience behaviour, royalties, and global reach. However, concerns remain regarding limited auditability, opaque payment mechanisms, and the power imbalance inherent in non-negotiable platform contracts. A key gap relates to how independent artists navigate these constraints compared with those represented by major labels. Ensuring transparency and equitable data access is therefore central to maintaining artist trust and strengthening the long-term viability of platforms like Spotify.

6 User Rights, Access Limitations and Ethical Experience

6.1 Ownership vs Access

Spotify users do not receive ownership of digital music; instead, they are granted only limited, revocable access such as listening to music on the platform with an active subscription. Even premium "offline mode" downloads are securely encrypted and tethered to the user's account and device, so when the subscription ends, decryption keys expire and downloaded files become inaccessible. Courts from the US and EU have ruled that these digital "copies" are not subject to the initial first-sale or exhaustion doctrines. As a result, secondary markets for digital music are blocked, and the user's relationship to music is transformed from property to conditional access.Srivastava, 2022 Scharf, 2022Matin, 2007

6.2 Subscription and Access Control

Spotify has different subscription plans: free users can only access content with ads, no offline music, and several restrictions, while premium users receive more privileges but are ultimately bound by DRM, which prevents copying, sharing, and even playback on unauthorized devices. This approach increases revenue extraction and control; however, it comes at a cost to users and their freedom of usage. Scharf, 2022

6.2.1 Privacy and Surveillance

Requests for every playback are recorded, including tracking time, device ID, IP address, and usage patterns for content licensing and anti-fraud purposes. However, this automated measurement is repurposed for sophisticated behavioural marketing. The ACS Principle of Privacy expresses that the collection of data should be minimal and transparent, but this requires detailed academic analysis because platform policies often change and opt-out systems are limited.Rafi et al., 2023Srivastava, 2022

6.3 Accessibility and Inclusion

Offline DRM verification excessively affects users who live in regions with patchy connectivity. DRM systems that lack universal or inclusive design can worsen digital divides. Researchers strongly support incorporating accessibility and inclusion principles into future DRM standards, emphasizing that digital technologies must optimize, not obstruct, and enhance users' quality of life.Srivastava, 2022

6.4 Synthesis

While discussions continue regarding the ethical justification of DRM restrictions, the literature consistently recognises that DRM tightly regulates playback, limits user ownership, and constrains fair use. Unanswered questions persist about the effects of DRM on users in low-connectivity environments and those with limited digital access. Privacy concerns arising from behavioural tracking also remain insufficiently examined within the context of music streaming. Addressing these issues is essential for Spotify to uphold user trust, foster inclusivity, and align its practices with broader digital rights and data-protection principles.

7 Global Legal Frameworks

The international legal authorization of DRM rests on treaties and national laws that reshape both creators' and users' rights in the digital era. The WIPO Copyright Treaty (WCT) of 1996, together with influential national statutes such as the U.S. DMCA and the EU Information Society Directive (InfoSoc), requires signatory nations to protect not only traditional copyright but also the technological protection measures (DRM) that enforce it. Regardless of user intent or other legal rights (such as fair use or personal backup), these laws extend far beyond conventional copyright enforcement by making it

illegal to distribute circumvention tools or to bypass DRM systems. Srivastava, 2022
As a result, users and even researchers are legally restricted in their ability to examine, question, or bypass DRM even when their intentions are legitimate. This creates a powerful legal and technical barrier that can suppress innovation and limit fair dealing.
This international legal framework gives platforms like Spotify strong, globally recognized authority. The platform's licensing model and DRM systems are reinforced by law across multiple jurisdictions, making it impossible for users to invoke secondary rights such as resale, duplication, or redistribution of music on other platforms. In practice, DRM-supported streaming ecosystems transform digital content into tightly controlled, access-based services, protected simultaneously by technology and international legal regulation.Srivastava, 2022

7.1 Synthesis

Researchers indicate that international laws such as the DMCA, WIPO treaties, and the EU InfoSoc Directive significantly reinforce DRM protections across jurisdictions. However, scholars disagree on whether these frameworks strike an appropriate balance between copyright enforcement and user freedoms or innovation. A major research gap concerns how these legal structures influence accessibility, competition, and long-term platform dominance. For Spotify, such legal frameworks legitimise and support DRM enforcement at a global scale, shaping licensing, distribution, and content control across diverse markets.

8 Platform Ownership , Artist Possession and Intermediation

8.1 Who Owns the Music? Artis Possession vs Platform Control

Spotify users do not receive ownership of digital music; instead, they are granted only limited, revocable access such as listening to music on the platform with an active subscription. Even premium “offline mode” downloads are securely encrypted and tethered to the user's account and device, so when the subscription ends, decryption keys expire and downloaded files become inaccessible. Courts from the US and EU have ruled that these digital “copies” are not subject to the initial first-sale or exhaustion doctrines. As a result, secondary markets for digital music are blocked, and the user's relationship to music is transformed from property to conditional access.Srivastava, 2022

8.2 Spotify as Third-Party Business Intermediary

In the music industry, Spotify serves as a powerful music digital platform which bringing together musicians and users while managing access, data, and payment terms. To influence visibility and profits, the business uses recommendation algorithms, negotiates with significant rightsholders (who could be stakeholders), and establishes payout guidelines. Because Spotify's legal and technical terms make it difficult for artists and users to work outside of or switch platforms without losing audiences, control, or content, this business model leads to "platform lock-in." Thus, platforms have continuous control over music distribution, income, and discovery rather than users or many creators.Zhang and Lou, 2011

8.3 Synthesis

The literature suggests that while artists maintain original copyright, contractual arrangements and DRM enable labels and platforms to exert greater practical control over distribution and commercial use. Debate persists on whether this model empowers creators through global reach or restricts them through contractual dependencies and platform governance. Limited research exists on how different types of artists, particularly independent creators, experience these intermediation structures. For Spotify, this continued tension underscores the importance of ensuring balance between platform authority and creator autonomy.

9 Ethical Reflection: Application of the ACS Code of Ethics to Spotify's DRM

The public interest, quality of life, honesty, and fairness—the fundamental principles of the ACS Code of Ethics must be balanced against the issues raised by Spotify's DRM system, even though it enables widespread legal access to music and protects copyright. By limiting user and artist access through permission-based controls, Spotify's DRM erodes genuine ownership, restricts personal and creative freedoms, and consolidates market power among major labels and platforms. These policies reduce user autonomy, make it harder for artists to understand their payouts, and hinder users' ability to give informed consent due to unclear and non-negotiable terms.Srivastava, 2022Matin, 2007

Musical diversity and equity are affected because artists have less control over how their work is distributed, and smaller or independent artists face greater obstacles to competition. According to the ACS, this falls short of promoting inclusivity and fairness. Furthermore, DRM limits acceptable secondary uses, such as accessibility or educational exceptions, and sometimes prioritises profit over the public good, even though it may reduce overall piracy and support corporate sustainability.Srivastava, 2022Matin, 2007

9.1 Synthesis

Studies emphasise that the extensive application of DRM on streaming platforms operates as a double-edged sword: it supports anti-piracy objectives and underpins modern subscription-based business models, yet it also creates structural disadvantages for users and creators by limiting ownership, fair use, and transparency. Scholars highlight the need for greater algorithmic openness, fair-use flexibility, accessibility, and clearer communication of rights to align DRM practices with ACS principles of fairness, public interest, inclusivity, and quality of life. For platforms like Spotify, adopting such measures would strengthen ethical alignment and support a more equitable digital environment.Zhang and Lou, 2011

10 Conclusion

This literature review has demonstrated that Spotify's DRM-centred business model has fundamentally reshaped the music industry, shifting it from an ownership-based system to a tightly regulated, platform-controlled access model. The review explored technical, legal, and ethical dimensions, showing how DRM enables creators and rightsholders to reduce piracy, ensure royalty payments, and reach global listenership. However, these benefits come at a cost: DRM entrenches platform dominance, restricts user and artist autonomy, and transforms music access into a series of time-limited, non-transferable permissions.Srivastava, 2022 Scharf, 2022 Zhang and Lou, 2011Beekhuyzen, 2006

The most profound challenges identified include the opacity of royalty distribution, the heavy concentration of power in the hands of platforms, the elimination of meaningful ownership and secondary markets for users, and unresolved concerns regarding fairness, inclusivity, and algorithmic transparency. For independent musicians and marginalized communities, the current DRM regime can compound vulnerabilities and intensify reliance on major industry gatekeepers.

From this research, future investigation and policy change should focus on:

- Enhancing transparency in royalty calculation and contractual terms.
- Expanding fair use and accessibility for users and creators.
- Auditing DRM algorithms for fairness and bias.
- Supporting independent and small-scale creators within platform economies.
- Ensuring DRM systems are subject to public oversight and designed with user rights and artist equity at their core.

Without these reforms, DRM risks undermining not only trust and market competition, but also equitable participation in digital culture. It is crucial for future technology and policy leaders to create DRM environments that balance copyright protection with the broader values of transparency, inclusion, and creative innovation.Srivastava, 2022 Scharf, 2022 Zhang and Lou, 2011Beekhuyzen, 2006

11 Appendix: Literature Review Preparation Process

The following outlines the initial draft structure of my literature review, presented using bullet points, section titles, and subtitles. This draft served as the organisational foundation for developing the final, fully written version of the review. It reflects how key themes, subtopics, and analytical areas were identified and categorised before expanding them into complete academic sections.

11.1 Introduction

11.1.1 Background and Context

- Shift from physical music ownership to streaming access, enabled by DRM.
- Spotify as an ideal case study due to its global influence and unique approach.
- DRM's dual function: securing content and raising ethical/policy debates.

11.1.2 Relevance to Information Security

- DRM's role in organizing permission-based access and protecting digital assets.
- Impact on digital trust, privacy, and compliance within platform ecosystems.

11.1.3 Research Aim and Questions

- How Spotify's DRM changes ownership, access, licensing, and rights relationships.
- Key questions on transparency, equity, and ethics for users and artists.

11.1.4 Article Selection Rationale

- Importance of copyright, technology, ethics, and market practice in digital music.
- Final selection: 8 core articles (many directly analyzing Spotify practices).

12 Body

12.1 Business Models and Company Ownership of DRM

12.1.1 Centralisation and Enforcement of Access

- Spotify's role in shifting from ownership to access-based licensing using DRM.
- DRM centralizes control, restricts user actions, and enables platform lock-in.

12.1.2 Licensing and Rights Possession

Record labels retain copyright; Spotify retains control via DRM and licensing.

Spotify manages encryption, playback, and distribution infrastructure.

12.1.3 Licensing and Royalty Frameworks

- Performance-based contracts, automated royalty tracking, and compliance.
- DRM ensures exclusive control and prevents unauthorized redistribution.

12.1.4 Spotify's Business Strategy with DRM

- Monetization through subscription tiers; DRM underpins service differentiation.
- Proprietary DRM increases platform's power over content, users, and partners.

12.1.5 Artist Protection, Revenue Integrity, and Ethical Tensions

DRM analytics protect royalties but raise transparency/fairness issues.

Ethical debates on concentration of power and exclusionary practices.

12.2 . DRM and Copyright Challenges in the Spotify Ecosystem

12.2.1 Erosion of User Rights and Market Ownership

- DRM transforms ownership into limited, revocable licenses.
- End-user license agreements further restrict user flexibility and secondary markets.

12.2.2 High-Profile Copyright Disputes

Taylor Swift catalogue removal as a case study in artist-platform conflict.

DRM's enforcement power can disadvantage artists and users alike.

12.3 Security and DRM-Related Incidents

12.3.1 DRM System Vulnerabilities and Threats

Technical weak points: Google Widevine vulnerabilities, exploit risks.

Academic/legal barriers to comprehensive DRM security research.

12.3.2 Songwriters' Royalty Disputes

Tension in DRM-based royalty tracking and disputes with publishers like Kobalt.

Dependence on platform-controlled validation.

12.4 Artist Rights and Access Control

12.4.1 Artist Access to DRM Data

- Spotify for Artists provides access to comprehensive streaming metrics.
- DRM-linked data supports targeted marketing and licensing strategies.

12.4.2 Benefits and Ethical Concerns

- Revenue tracking and piracy prevention as key DRM benefits.
-
- Lack of independent audits, opaque payment algorithms, and algorithmic decision-making power centralization.
-
- Need for open audit practices and equitable contract negotiation.

12.5 User Rights, Access, and Ethical Experience

12.5.1 Ownership vs. Access

Spotify users get access, not ownership; even offline downloads remain under DRM control.

12.5.2 Subscription and Access Control

DRM-enforced function differentiation between free and premium users.

12.5.3 Privacy and Surveillance

All playback and behavioral data is collected for licensing—and marketing—purposes.

12.5.4 Accessibility and Inclusion

Offline DRM can exclude users with poor connectivity; calls for universal design.

12.6 Global Legal Frameworks

12.6.1 International Legal Authorization

- WIPO, DMCA, and EU InfoSoc Directive enforce DRM protections.
- Global frameworks reinforce platform control and limit user research/fair use

12.7 Platform Ownership, Artist Possession, and Intermediation

12.7.1 Who Owns the Music?

Artists may retain copyright, but practical power is in hands of platforms and labels via DRM.

12.7.2 Spotify as Business Intermediary

'Platform lock-in' preserves Spotify's control over content, revenue, and discovery.

12.8 Ethical Reflection

12.8.1 Application of ACS Code of Ethics

12.9 Conclusion

Key Findings and Recommendations

Spotify's DRM shifts industry structure, secures content, but raises major concerns regarding fairness, transparency, platform power, and artist/user autonomy.

Recommends policy changes for transparency, auditability, equity, and public oversight.

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